http://www.diva-portal.org

This is the published version of a paper published in *Annals of Medicine*.

Citation for the original published paper (version of record):

Odzakovic, E., Anefur, E., Oberg, S., Jakobsson, M., Bjork, M. et al. (2026)
Biopsychosocial barriers and facilitators in the everyday lives of patients with restless legs syndrome: a qualitative study
Annals of Medicine, 58(1): 2743219
https://doi.org/10.1080/07853890.2026.2743219

Access to the published version may require subscription.

N.B. When citing this work, cite the original published paper.

Permanent link to this version:
http://urn.kb.se/resolve?urn=urn:nbn:se:lnu:diva-150328

Annals of Medicine

ISSN: 0785-3890 (Print) 1365-2060 (Online) Journal homepage: www.tandfonline.com/journals/iann20

Biopsychosocial barriers and facilitators in the everyday lives of patients with restless legs syndrome: a qualitative study

Elzana Odzakovic, Elin Anefur , Sandra Öberg, Malin Jakobsson, Maria Björk, Susanne Knutsson, Lise-Lotte Jonasson , Mattias Georgsson , Martin Ulander, Bengt Fridlund, Jonas Lind & Anders Broström

To cite this article: Elzana Odzakovic, Elin Anefur , Sandra Öberg, Malin Jakobsson, Maria Björk, Susanne Knutsson, Lise-Lotte Jonasson , Mattias Georgsson , Martin Ulander, Bengt Fridlund, Jonas Lind & Anders Broström (2026) Biopsychosocial barriers and facilitators in the everyday lives of patients with restless legs syndrome: a qualitative study, Annals of Medicine, 58:1, 2743219, DOI: 10.1080/07853890.2026.2743219

To link to this article: https://doi.org/10.1080/07853890.2026.2743219

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 13 Oct 2026.

Submit your article to this journal

Article views: 366

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=iann20

ANNALS OF MEDICINE
2026, VOL. 58, NO. 1, 2743219
https://doi.org/10.1080/07853890.2026.2743219

RESEARCH ARTICLE

OPEN ACCESS Check for updates

Biopsychosocial barriers and facilitators in the everyday lives of patients with restless legs syndrome: a qualitative study

Elzana Odzakovic[a], Elin Anefur[a], Sandra Öberg[a], Malin Jakobsson[a], Maria Björk[a], Susanne Knutsson[b,c], Lise-Lotte Jonasson[a], Mattias Georgsson[a], Martin Ulander[d,e], Bengt Fridlund[c], Jonas Lind[e,f] and Anders Broström[a,d,g]

[a]Department of Nursing, Jönköping University Halsohogskolan, Jönköping, Jonkoping County, Sweden; [b]Department of Health and Caring Sciences, Faculty of Health and Life Sciences, Linnaeus University, Växjö, Sweden; [c]Centre for Interprofessional Collaboration within Emergency Care (CICE), Linnaeus University, Växjö, Sweden; [d]Department of Clinical Neurophysiology, Linköping University Hospital, Linköping, Sweden; [e]Department of Biomedical and Clinical Sciences, Division of Neurobiology, Linköping University, Linköping, Sweden; [f]Section of Neurology, Department of Internal Medicine, County Hospital Ryhov, Jönköping, Sweden; [g]Department of Health and Caring Sciences, Western Norway University of Applied Sciences, Bergen, Vestlandet, Norway

ABSTRACT

Background: Restless legs syndrome (RLS) is a common condition affecting 3% of the world's population. Pharmacological treatments do not cure RLS, and symptoms substantially affect everyday life. However, knowledge of biopsychosocial (BPS) factors influencing patients' everyday lives remains limited. This study aimed to explore and describe BPS barriers and facilitators influencing everyday life among patients with RLS.

Methods: A deductive, descriptive qualitative design grounded in the BPS model was employed. Semi-structured telephone interviews were conducted with 28 strategically selected patients from a national RLS organisation. Data were analysed using qualitative content analysis and structured according to biological, psychological, and social barriers and facilitators.

Results: Biological barriers included RLS-related symptoms, sleep disturbances, and fatigue, while physical activity and environmental adaptations facilitated symptom management. Psychological barriers involved despair and inadequate healthcare responses, whereas access to information and supportive healthcare interactions enhanced everyday life. Social barriers included limitations in daily activities and the need for lifestyle adjustments. Support from patient organisations and understanding social networks facilitated coping with RLS. Together, these findings highlight how interconnected BPS factors shape patients' everyday experiences and ability to manage their condition.

Conclusions: Comprehensive RLS care should address biological, psychological, and social barriers, particularly sleep disturbances and care-related stress, while promoting facilitating factors such as lifestyle adaptations and social support. Future research should evaluate the effects of comprehensive care interventions on symptom severity and overall BPS functioning in RLS.

ARTICLE HISTORY
Received 5 June 2026
Accepted 24 September 2026

KEYWORDS
Biopsychosocial approach; patients; sleep; support; qualitative research; Willis–Ekbom disease

Background

Restless legs syndrome (RLS) is a common, chronic, sleep-related sensorimotor disorder [1] affecting nearly 3% of the global population [2]. RLS is typically diagnosed in primary care using five established criteria: an urge to move the legs, worsening at rest, relief with movement, nighttime predominance, and exclusion of other conditions [1]. Despite these criteria, underdiagnosis remains common, as non-specific symptoms, such as mood disturbances [3,4] or leg pain [5], often overlap with other conditions and may be difficult to describe [1]. The biological core of RLS is characterised by a nocturnal urge to move and physical discomfort, such as pain or unpleasant sensations in the legs. However, the impact of the disorder extends

CONTACT Elzana Odzakovic elzana.odzakovic@ju.se Department of Nursing, Jönköping University Halsohogskolan, Jönköping, Jonkoping County, Sweden.
© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

beyond these primary symptoms and may affect several aspects of everyday life. Sleep problems are common and may occur as a consequence of the symptoms [6,7], while the fluctuating and unpredictable nature of RLS can contribute to psychological distress and emotional strain [6,8]. These challenges may also result in social limitations and disrupt occupational functioning and everyday activities [9].

Individualised pharmacological treatment, such as alpha-2-delta ligands, is often recommended following assessment of comorbidities and iron status [5,10,11]. However, pharmacological treatments cannot cure the condition and may be difficult to optimise. Treatment may fail to provide complete symptom relief and, in some cases, may exacerbate symptoms through augmentation, negatively affecting everyday life [12] and overall quality of life [13]. Therefore, symptom management may also involve self-care strategies [12,14,15]. These challenges highlight the need for clinical assessments and care approaches that extend beyond biological symptoms and consider the broader consequences of living with RLS.

The biopsychosocial (BPS) model (Figure 1) provides a framework for understanding health and illness through interactions among biological, psychological, and social factors [16,17]. According to Engel [16], the biological factor includes physiological and pathological processes, which, in the context of RLS, may involve the condition's various physical manifestations. The psychological factor encompasses cognitive and emotional states, including distress, fear-avoidance, and learning processes. The social factor includes broader environmental and cultural influences, such as socioeconomic circumstances and social support networks.

From a BPS perspective, the biological symptoms of RLS may interact with psychological and social factors, thereby influencing how patients experience and manage their everyday lives. For example, the circadian symptom profile and the risk of augmentation may affect sleep and daily functioning, while psychological distress and social circumstances may further influence the individual's ability to manage the condition [6,9]. Engel highlighted the importance of considering biological, psychological, and social factors, as life circumstances can affect disease progression, symptom persistence, and the relationship between the patient and healthcare professional [17,18]. Thus, adopting a BPS perspective may contribute to a more comprehensive understanding of the complex interactions that influence everyday life with RLS. Although the BPS model has been applied to other chronic conditions, such as chronic pain [19], Parkinson's disease [20], and depression [21], it has not yet been employed in an RLS context. The dynamic interplay between biological, psychological, and social aspects of RLS and their influence on everyday life therefore remains insufficiently described.

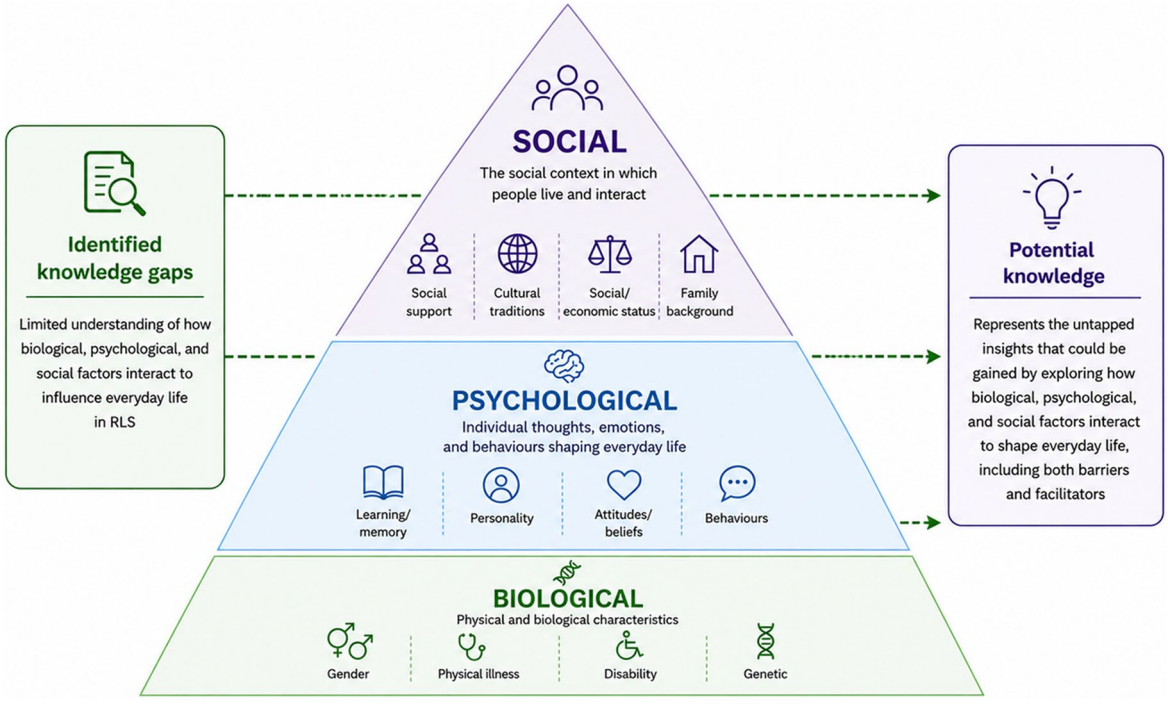


Figure 1. Description of the three factors included in the biopsychosocial model.

By identifying barriers and facilitators from a BPS perspective, healthcare professionals may gain a broader understanding of how patients manage everyday life with RLS and the factors that influence their experiences. Therefore, this study aimed to explore and describe BPS barriers and facilitators influencing the everyday lives of patients with RLS.

Methods

Study design and participants

A deductive descriptive design with qualitative content analysis [22] was employed based on the BPS model [16]. Participants were recruited from the Swedish RLS Patient Association. Approximately 1500 invitation letters were distributed *via* a questionnaire survey, resulting in 788 participants, as detailed in Hellström et al. [23]. All patients were asked about their willingness to be interviewed, and 472 consented. Background information from the questionnaires was utilised for strategic sampling [24]. Strategic sampling ensured variation in sociodemographic and clinical characteristics within the study population, capturing a broad range of experiences and self-care practices related to living with RLS. Of the 472 participants who had consented to be contacted for potential participation in the interview study, 28 participants were strategically selected based on gender, age, educational level, cohabitation status, comorbidities, RLS symptom severity, and pharmacological treatment. The selection criteria were informed by the interdisciplinary research team's theoretical knowledge and clinical expertise regarding the diverse manifestations of RLS across different contexts and disease trajectories. The inclusion criteria were being 18 years of age or older, having a confirmed diagnosis of RLS and ongoing treatment, and being able to speak and understand Swedish. All 28 selected participants consented to participate in the interview study. Table 1 summarizes the participants' demographic and clinical characteristics.

Data collection

The data collection was conducted via telephone interviews between June and November 2022. A semi-structured interview guide (Table 2) was designed by the interdisciplinary research team, consisting of physicians, nurses, and sociologists who had extensive experience in treating patients with RLS and

Table 1. Demographic data of the patients with RLS ($N = 28$).

Variables	Value
Gender, female, *n* (%)	16 (57)
Age (years), mean (range)	67,6 (39–89)
Educational level, *n* (%)	
9 years or below	3 (11)
12–13 years	11 (39)
University	14 (50)
Civil status, *n* (%)	
Married/living together	20 (71)
Unmarried and living alone	3 (11)
Divorced/widower and living alone	5 (18)
Smoking, *n* (%)	
Yes, *n* (%)	1 (4)
Alcohol, *n* (%)	
Never uses alcohol	8 (29)
Uses alcohol 2–3 times or more/week	11 (39)
Comorbidity, *n* (%)	
Renal disease	0 (0)
Parkinson's disease	0 (0)
Multiple sclerosis	0 (0)
Migraine	3 (11)
Iron deficiency	4 (14)
Pharmacological treatment, *n* (%)	
Dopamine agonists	22 (80)
Opioids	8 (29)
α2δ Ligands	7 (25)
Dopa/derivates	5 (20)
Iron supplement	3 (11)

conducting qualitative studies. The semi-structured interview guide contained eight questions focusing on factors such as life situation and symptoms during the day and nighttime. Probing questions were used to deepen and clarify the issue during the semi-structured interview [25]. Two pilot interviews were conducted to refine the semi-structured interview guide, ensuring that the questions generated rich and relevant data. This process also allowed for adjustments to improve clarity and validity [25]. The pilot interviews demonstrated that the interview guide was relevant and required no changes. The participants choose the date and time for the semi-structured interview. The interviews, which lasted 45–90 min, were audio-recorded and transcribed verbatim by a transcriptionist.

Data analysis

Transcripts from the 28 semi-structured interviews were produced, resulting in 314 A4 pages. The deductive qualitative content analysis was based on the Graneheim and Lundman method [22] and the three factors of the BPS model (Figure 1). First, all the text was read several times to identify statements (i.e. meaning units) concerning the three factors of the BPS model (i.e. biological, psychological and social) [16]. Secondly, the meaning units were read repeatedly, condensed and coded, and then compared and deductively clustered into the three factors. Thirdly, the interdisciplinary research team engaged in discussions to establish a consensus on a category system describing barriers and facilitators for each of the three factors in the BPS model. The system included the categories and subcategories demonstrated in Figures 2–4 and the text in the results section describing barriers and facilitators affecting the everyday lives of patients with RLS. The entire interdisciplinary research team participated in the analysis, engaging in ongoing discussions to challenge interpretations, refine category boundaries, and enhance the study's trustworthiness. Any disagreements or uncertainties were resolved through reflexive dialogue until a shared understanding was reached.

Ethical considerations

The study was approved by the Swedish Ethical Review Authority in Stockholm, Department 4 Medicine (reference: 2022-01515-01), and was conducted in accordance with the principles outlined in the Declaration of Helsinki [26]. Independent ethical review was obtained from an independent ethics review authority in accordance with Swedish guidelines and regulations for research. Approval was sought from an independent review body because the study was conducted outside the framework of a specific institutional ethics committee. This approach ensured that the ethical assessment was performed impartially and in compliance with established Swedish standards for research ethics. The participants were informed through an information letter that the research was a scientific study involving semi-structured interviews. They were assured that their real names would not be used in the data analysis; instead, each participant was assigned a code number (Informant 1–28). Written informed consent was obtained from all participants, who were required to return the signed consent form to be included in the study.

Table 2. Examples of questions from the semi-structured interview guide.

Examples of questions	Examples of probing questions
Would you please share your experiences of what a typical day might look for you?	Could you please provide further explanation on the occurrence and experiences of symptoms through the day, fatigue and their effect on mood?
Would you please share your experiences of what a typical night might look for you?	Can you provide further explanation regarding the occurrence and experiences of nocturnal symptoms and how they affect your sleep?
Can you describe how you experiences your life situation with RLS?	Can you explain that further relate to your work, family, social life, health, lifestyle and living conditions?
Can you describe how you perceive the support that has been given to you?	Can you provide further explanation about the specifics of the support you have received, including how it was given, when it was given, and the quality of the support you have received?

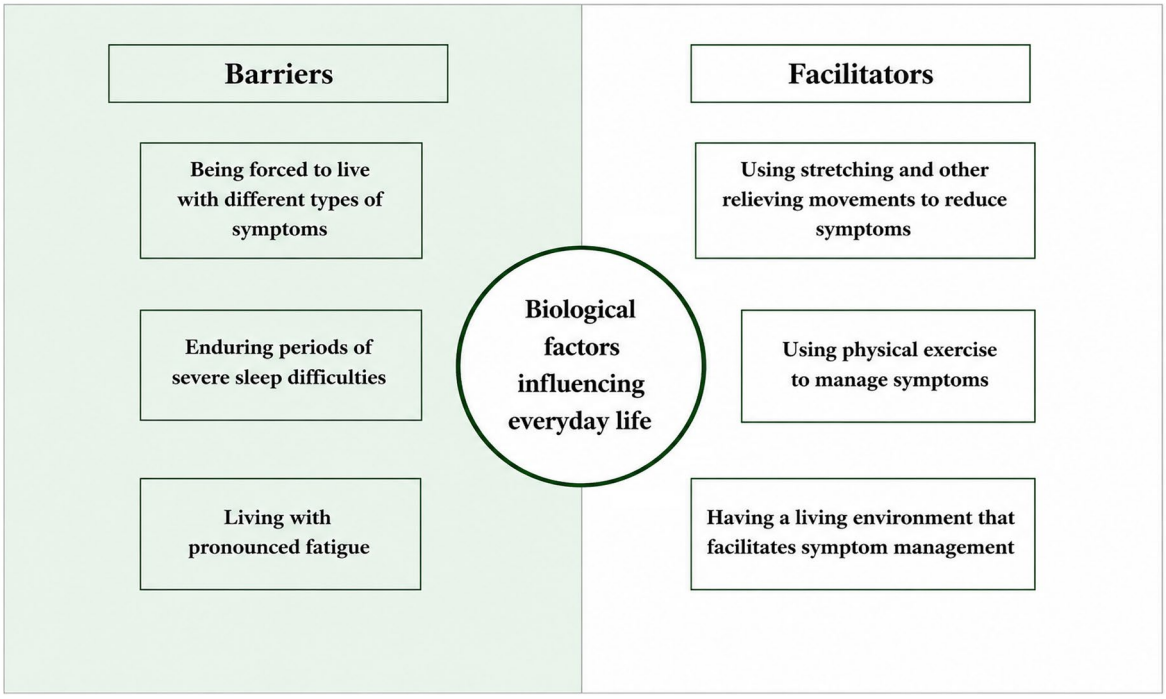


Figure 2. Biological barriers and facilitators for everyday life among patients with restless legs syndrome.

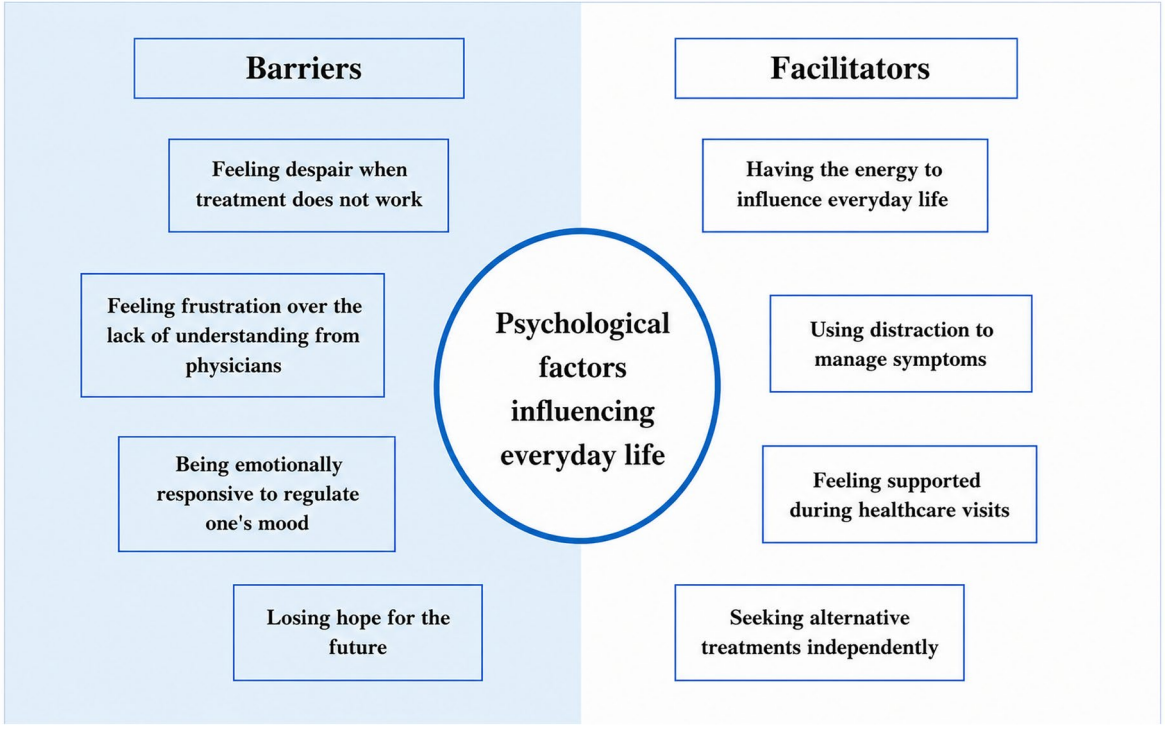


Figure 3. Psychological barriers and facilitators for everyday life among patients with restless legs syndrome.

Trustworthiness

This study followed the four criteria of scientific trustworthiness, which included credibility, transferability, dependability, and confirmability [26]. Credibility and confirmability were established by recruiting only those participants who met the inclusion criteria and by using direct quotations from all participants' statements. A comprehensive description of all methodological processes used in this study, along with

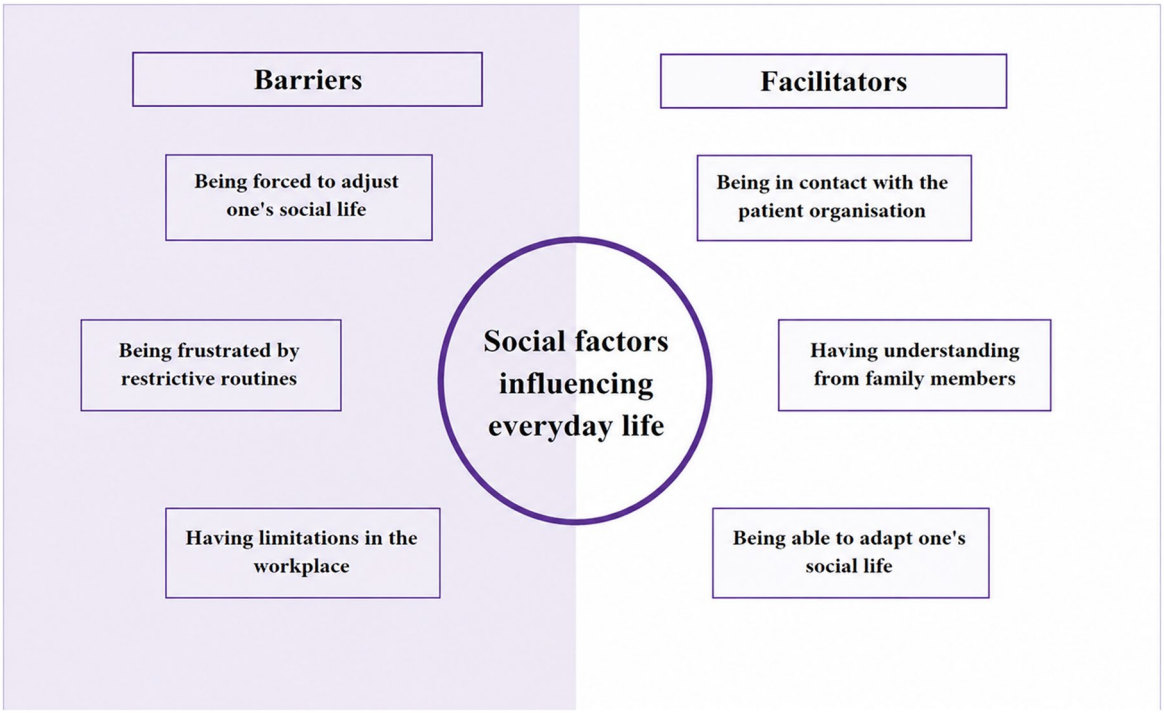


Figure 4. Social barriers and facilitators for everyday life among patients with restless legs syndrome.

the study's context, has been provided in the method section and the limitations, which enhances the credibility, dependability and transferability of the research.

Results

This study involved 28 participants aged 39–89 years, including 16 women (57%) and 12 men, the majority of whom held a university-level education. Detailed demographic characteristics are presented in Table 1. The examination of data on BPS barriers and facilitators impacting the everyday lives of patients with RLS identified three main categories, each containing several subcategories (Figures 2–4).

Biological factors influencing everyday life with RLS (Figure 2)

Biological barriers

Being forced to live with different types of symptoms. Symptoms were unpredictable, occurring in periods, like "attacks", or all the time. They were described as discomfort and pain in the legs and other body areas, mainly starting during inactivity. In some cases, when the symptoms were mild, they felt like exercise soreness or muscle tension; when more intense, they felt like cramps, with tingling and crawling sensations. The pain ranged from irritation and discomfort to burning or stabbing sensations. The feet felt icy cold but were warm to the touch. Others described feeling like their legs were boiling, and they wanted to ease the feeling by crawling out of their own skin. Symptoms often worsened over time, and when they manifested, a sense of going crazy could be experienced.

> *"Once it begins, I experience it in both legs and arms. Often it persists for almost the entire day and can last for several weeks at a time".* Informant # 8.

Enduring periods of severe sleep difficulties. Sleep difficulties often formed a vicious cycle, with poor sleep causing further disturbances, making regular sleep a rare occurrence. Anticipating awakenings due to leg sensations or pain also created stress and anxiety at bedtime. Difficulties falling asleep were described, even when feeling extremely tired, like "a zombie". Severe RLS symptoms, described as episodes with a "crawling sensation", could occur during the night, shortening sleep. Once the symptoms started in the evening or night, they were difficult to alleviate with medication. Some patients described sleeping

separately as a way to reduce stress and allow greater freedom to move without disturbing others. During severe RLS symptoms and difficulties sleeping at night, patients tried alternative sleeping arrangements and often had to walk extensively indoors at night.

> "*After I've woken up and been up for a while and intend to go to bed again, that's when I have a lot of discomfort in my legs, that's when I can't lie in bed, but then just realise that I can't lie here now, I have to stand up and walk, I need to move around*". Informant # 15.

Living with pronounced tiredness. The pronounced tiredness affected everyday life and, among other things, made it difficult to stay awake when inactive. If still at work, it was described as a struggle not to fall asleep while participating during meetings and breaks. Tiredness caused listlessness, reduced motivation, and difficulties with thinking, memory, concentration, and decision-making. Short rests provided only temporary relief. RLS symptoms could occur in the morning and intensify with daytime tiredness, leading patients to avoid driving when exhausted.

> "*The feeling is stronger than tiredness… when you don't get any sleep, and if it's been several days too, then it's no fun. So that uh, I'm struggling with this. That, that's tricky. It's like torture, almost*." Informant # 17.

Biological facilitators

Using stretching and other relieving movements to reduce symptoms. In general, feelings of pain and discomfort were difficult to control. Sitting still when experiencing symptoms was impossible because the situation required relieving movements. Leg muscle stretching exercises were described as providing a sense of comfort without triggering symptoms. Specific forms of movement causing "another sort of pain" were used since they were perceived to decrease RLS-related pain. Stomping, kicking with the legs, moving and waving the arms were also used to reduce symptoms, even at night.

> "*The only thing I think helps, as I said, if the problems come at night, is to stretch and refuse to give up. It usually helps.*" Informant # 13.

Using physical exercise to manage symptoms. RLS symptoms were managed through physical exercise, adapted to the individual symptoms, circumstances, and capacity. Exercise was either performed at a particular time to prevent symptoms or to decrease existing symptoms. The importance of maintaining a balance between these two approaches was highlighted to get the best results. Being active during the day was important, as days with low physical activity were associated with more intense RLS symptoms and lower mood. Having a treadmill or exercise bike at home made it possible to be physically active also at night, which decreased pain and cramps and helped muscles relax.

> "*As soon as you move your legs in one way or another… one day I'm walking, another I'm pedalling the bike… well, then those crawling problems are relieved.*" *Informant* # 1.

Having a physical living environment that facilitates symptom management. To reduce symptoms, it was important that the accommodation enabled and facilitated symptom management. Access to an adequate living space, allowing movement at night when symptoms occurred, facilitated symptom management. The living environment, with several rooms, provided space to walk when RLS symptoms occurred; however, indoor stairs were considered difficult to climb, especially for older patients. Hot weather was perceived as a contributing factor to increased symptoms. A garden was described as beneficial because it facilitated movement, thereby facilitating symptom management.

> "*I'm living alone without neighbours, so I can walk around and do what I want. When I need to be up during the night, then I have somewhere to go without disturbing anyone, I can stay in the living room or in the kitchen*". Informant # 7.

Psychological factors influencing everyday life with RLS (Figure 3)

Psychological barriers

Feeling despair when treatment does not work. A significant psychological barrier was the feeling of despair when prescribed medication or other treatments, such as self-care, failed to ease symptoms.

Symptoms continued despite adherence, eroded confidence in the treatment and raised concerns about its future effectiveness. This led to ongoing uncertainty about how to manage the condition. Food intake was also seen as affecting medication efficacy, since the drugs did not consistently produce the intended results, which further heightened feelings of unpredictability and a loss of control. Despite extensive efforts to seek help, including visits to pain clinics and neurologists, as well as trying various non-drug approaches, such as acupuncture, patients reported minimal or no long-term improvement.

Self-care strategies were inconsistently effective, with variable responses increasing feelings of confusion, helplessness, and despair and highlighting the psychological burden of the condition.

> "*Even when I followed the treatment, the symptoms still came back. It made me feel hopeless, not knowing what would help or what to do next. Sometimes things worked, sometimes they didn't, and that uncertainty was really difficult to handle". Informant* # 22.

Feeling frustrated over the lack of understanding from physicians. A lack of understanding among physicians (e.g. general practitioners and specialist neurologists), characterised by a poor overall understanding of RLS, led to frustration. Poor commitment from physicians when communicating symptoms was perceived as a possible reason no further investigations were planned, or a diagnosis was not established. During care visits, physicians attempted to find suitable medication; however, the patients felt abandoned because no regular follow-up was scheduled. The experience of being poorly treated and sometimes having more knowledge of RLS than the physicians led to decreased trust. For example, a concern was described about having surgery or falling ill in the future and not being able to communicate and make oneself understood about RLS-related problems. Obtaining the correct diagnosis and receiving treatment were often delayed or missing, while other treatment alternatives were tried instead.

> "*They interpret everything as mental. The last time was maybe three months ago, when I talked about my symptoms, a doctor said outright, "This is mental," And I tried to explain that it is not at all.*" Informant # 14.

Being emotionally responsive to regulate one's mood. Sadness and depressive symptoms were common and described as being worsened by severe RLS symptoms and poor sleep, but also by how the related daytime sleepiness and fatigue affected the life situation. Patience with people, and control of emotions in different situations, both at home and at work, were also affected by how the illness manifested itself. Being constantly tired meant the patients felt irritated and stressed, and were easily agitated, which could manifest as outbursts of anger in inappropriate situations.

> "*There are so many reasons, there are many things that affect how I feel and how my mood is, and you can become impatient and irritated and emotional, and, for example, take offence at me because someone happens to say something. It is probably this tolerance limit that is being lowered.*" Informant # 15.

Losing hope for the future. Conquering everyday life with a burdensome and incurable disease, which affected the whole life situation, was difficult and stressful, as symptom-relieving treatment was difficult to establish, but also because support to deal with the situation was absent. Observing that some family members experienced more severe symptoms, the patients also feared that their own disease might progress similarly. It emerged that when symptoms were at their worst, participants found it difficult to make life work and experienced an inability to cope with anything. It was hard to find happiness, and the participants could only summon enough energy to do their work. When the symptoms occasionally increased, they felt they were in a black hole or a downward spiral, where it was difficult to keep fighting. When experiencing a worsened situation over the years, the patients felt broken down, since the disease was perceived as a torment.

> "*When there are nights like that (with severe RLS symptoms), if there are several nights, then I get so tired, I think life is a plague, I have almost no desire to live, I get depressed. Then I sit on the toilet and cry for a while and try to see if it gets better, but it doesn't". Informant* # 20.

Psychological facilitators

Having the energy to influence everyday life. Having the strength and ability to seek information was described as an important psychological facilitator, as practitioners often did not provide it. Participants

took the initiative to obtain new information in order to find treatment alternatives that could help them manage their situation. General knowledge of RLS as a condition, as well as advice and research-based knowledge, was obtained from patient associations or, in some cases, from scientific papers. Information and questions about medicines were sought at pharmacies and from the Swedish Medicines Products Agency. Participants sought private care to monitor ferritin and adjust treatment, including self-referrals to neurologists for intravenous iron infusions. Moreover, participants tried adjusting medications themselves as consultations were not performed frequently.

"*I have my symptoms 24 hours a day, more or less. I'm on strong medication and I've learned to deal with it in some way*". Informant # 18.

Using distraction to manage symptoms. Distractions, often based on personal interests, were used to avoid focusing on symptoms. For example, listening to or playing music was described as a way to lock symptoms out and put the problems on hold. Painting was another way to create a distraction as the person ended up in their own world, where the symptoms were forgotten. Moreover, carpentry or embroidery were other hobbies that decreased symptoms. Listening to audiobooks or podcasts, reading, knitting, or ironing were additional ways to shift the focus to something else. Multitasking, e.g. doing crosswords while the legs were in motion, was also used. Some used breathing techniques, while others tried eating, or scratching and tearing at their legs as distractions.

"*When I get symptoms, it's impossible to be still, then I go into the study and start playing a little, then you think about something else*". Informant # 7.

Feeling supported during healthcare visits. Receiving support during healthcare visits, such as having a physician who understood their problems, prescribed medications, and planned regular follow-up visits, was important. Being confirmed when they provided the physician with RLS-related information (e.g. articles, reports, or books) led to feeling involved and noticed. The feeling of support was also strengthened when the participants themselves were allowed to propose and test some new drug, even though the doctor had other suggestions. Receiving the proper medication that helped to reduce severe symptoms was described as an "absolute lifesaver". On the other hand, when prescribed drugs did not work, the support from multi-professional teams was described as helpful to regain a good life.

"*She (the doctor) came up with the idea, 'I'll give you some pramipexole [Dopamine agonists] and we'll probably find a good solution. It felt much better, and then we agreed that it must continue that way*." Informant # 16.

Seeking alternative treatments independently. Non-pharmacological alternatives were sought when medication did not work or adverse effects occurred. Food supplements, reflexology and healing, as well as vibration machines (devices featuring a plate designed to be stood upon) and magnetic belts that could be tightened around the calf, were used for mild symptoms. Cupping, which involved pressing rubber cups onto the thigh to relieve leg pain, was another alternative tried. Flushing the legs with hot or cold water or placing the feet in the snow or on a concrete slab were other self-care actions used. Massaging the thighs and calves was sometimes described as helpful in reducing symptoms, along with medication.

"*Then I tried a kind of massage gun, one that vibrates. I run it over the muscles of the calves.*" *Informant* # 17.

Social factors influencing everyday life with RLS (Figure 4)

Social barriers

Being forced to adjust one's social life. Social adjustments were often needed based on how bothersome the symptoms were. In general, sedentary activities, such as going to a movie or a play, or taking part in conferences or lectures, were strenuous, especially for those with severe RLS. Medication had to be planned, and RLS-related sleep disturbances could lead to sleepiness, increasing the risk of falling asleep during the day. Spontaneous visits by friends were not appreciated due to tiredness or feeling listless. On the other hand, planned meetings or parties were also difficult due to the uncertainty linked to

symptoms, which was difficult for friends to understand as the disease was not "visible". Holiday trips were sometimes difficult, as sleeping away was challenging when symptoms occurred at night, requiring the participants to get up and move, which meant waking other people. Longer journeys had to be cut short due to problems sitting still, but also because they felt it was unsafe to drive when sleepy. Moreover, air travel was described as difficult due to difficulties standing up and walking around.

> "*I've always thought it was fun to go out and meet people and sit in restaurants and drink wine and stuff like that, hang out, and I have to skip that. Going to the cinema is basically impossible. And being in a restaurant for more than 1½ hours, it doesn't work; then it starts to creep up, especially if you drink wine or beer, because then the symptoms come on like clockwork*." Informant # 7.

Being frustrated by restrictive routines. RLS forced patients to use strict routines in everyday life, requiring a strict schedule. It was imperative to plan the medication, sometimes even minute by minute, and then to be adherent to avoid worsening symptoms. The importance of going to bed at a fixed time and falling asleep before the symptoms appear was described as essential for obtaining several hours of sleep. However, a more extended period of severe symptoms meant breakfast had to be at lunchtime to obtain a few hours of sleep in the morning. Planning any activity was difficult and had to be avoided, as fatigue was at its worst at such times.

> "*I can't drive after 4 p.m. Because then, then if I have to go a longer distance, I can probably go 5 minutes to the store or so, but if I must go a longer distance….that, that's not possible, because I have to stop every five minutes. I have to get out and jump about because I can't sit still, I'm crawling out of my skin*." Informant # 4.

Having limitations in the workplace. Several work-related challenges were linked to the disease. For example, excessive sleepiness in the morning after a night of severe symptoms made it necessary to sleep in the mornings. This required flexitime, which in turn meant having to stay at work until late in the afternoon. Several consecutive bad nights required sick leave. If flexible work hours or leave due to illness were not enough, working part-time or taking a disability pension were the only alternatives, which harmed finances. Even if the idea was to retain permanent employment, a risk of dismissal was described in situations where understanding of the problems created by the disease was lacking.

> "*If it's really bad… then I can't work. I must be able to work…. my work involves using my head… and my head does not work when the symptoms are at their worst". Informant* # 8.

Social facilitators

Being in contact with the patient organisation. The patient organisation was described as a community where people with RLS provided digital and telephone-based emotional, informational, and practical support for managing care-related contacts, including lay experiences with medication. The organisation also published a member magazine, released books, monitored a Facebook group, and organised lectures on RLS-related topics. Ongoing research on various medical aspects of RLS (e.g. practical self-care activities and the importance of tracking one's ferritin levels) was also provided.

> "*I'm talking about sleep hygiene; it's important to be very careful about going to bed at a certain time in the evening and getting up. I must not disturb those times, or it will be much worse. I'm very addicted to routines, huh." Informant* # 17.

Having understanding from family members. Having an understanding family member who could provide support was important, especially when symptoms were difficult to manage. When the family members themselves had RLS, the patients felt they could identify with their symptoms, and more easily provide emotional support. Older relatives also provided information on how to contact the patient organisation to learn more about the disease. Family members who understood the daily variation of RLS symptoms planned dinners or other activities in the middle of the day. Moreover, they did not find it strange that the situation required the person to stand up or take a walk during dinner or other sedentary activity.

> "*They're good friends, so no one cares if I get up at a dinner and stand for a while*." Informant # 16.

Being able to adapt one's social life. Patients with RLS described strategies that helped them stay socially active despite changing symptoms. Key strategies included adjusting routines, listening to their bodies,

maintaining regular sleep, and planning to rest earlier in the day to prevent symptoms from worsening. During social occasions, they used practical approaches, such as standing or moving while watching TV, taking on chores like clearing dishes, or choosing seats near exits to stay mobile. Dietary changes, such as reducing sugar, coffee, and alcohol intake, or following a Mediterranean diet, were reported to help manage symptoms. Flexible work hours and carrying medication at all times were additional aids, allowing proactive steps to be taken, along with adaptation to symptom changes. These strategies collectively served as social facilitators, supporting participation and engagement despite the challenges caused by RLS.

> *"I am constantly looking for information. I go to the RLS association's website, and I read their magazine. It's very good to be part of an association like this, you know. I joined 5 years ago, and it's the best thing I've done. In this way, I hear about other people's experiences as well".* Informant # 17.

Discussion

Our results indicate that the participants in this study encounter a variety of biological, psychological, and social barriers that significantly affect everyday life. These barriers include unpredictable symptoms, sleep disturbances, tiredness, emotional strain, and social limitations. However, patients also employ various facilitators to manage their condition, such as engaging in physical activity, accessing supportive environments, practising self-care, and utilising resources from healthcare professionals or patient organisations. Despite these challenges, patients demonstrate resilience by proactively adjusting routines, seeking information and support, and integrating pharmacological and non-pharmacological strategies to improve well-being. This study extends previous research by providing a comprehensive understanding of BPS-related barriers and facilitators in RLS.

Biological factors influencing everyday life with RLS

Biological barriers operated through interactions between core sensory and motor symptoms, disrupted sleep, and persistent tiredness. These processes generated distressing sensations, limited restorative sleep, and impaired daytime functioning. The participants in our study had heterogeneous symptom profiles consistent with evidence indicating multifactorial mechanisms, including genetic susceptibility and altered neural processing [27]. Brain iron deficiency, a key pathophysiological factor, affects dopaminergic function and sensory regulation, providing a substrate for symptom persistence [11,28]. Sleep disturbance emerged as a central barrier, to restorative sleep reflecting direct interference from RLS symptoms and secondary psychological and behavioural responses, such as difficulty initiating and maintaining sleep due to crawling or tingling sensations, and muscle tension [6,7]. Previous research shows that women more frequently report severe sensory symptoms and greater sleep disruption, potentially intensifying tiredness [10]. These findings underscore the need to consider individual variability when assessing the impact of RLS and planning interventions.

The BPS-related barriers and facilitators mentioned above, and the way they are described and understood in clinical settings, can be framed by the 3P model of insomnia. This model illustrates how predisposing factors, such as genetic and neurobiological vulnerabilities, interact with precipitating factors like symptom onset, and perpetuating factors, including conditioned arousal and maladaptive coping, to maintain the disorder [29,30]. The model is supported by polysomnographic studies, including EEG, showing altered sleep architecture, increased micro-arousals, and heightened physiological activation in RLS [31,32]. Symptom fluctuation and treatment-related augmentation may reinforce these barriers, making management challenging. Persistent tiredness results from continuous nocturnal disruption, heightened physiological arousal, and ongoing symptom burden, contributing to cognitive and physical impairment, reduced work capacity, and limitations in everyday life. Clinically, these findings support comprehensive, individualised management targeting both symptom reduction and broader biological barriers. Pharmacological treatment combined with iron assessment is recommended [14], while intravenous iron and non-pharmacological interventions such as tonic motor activation (TOMAC) reduce symptoms and improve sleep [33] in an RLS context.

Patients also actively use biological facilitators to mitigate RLS symptoms. Physical activity, including walking, running, and stretching, helps manage discomfort, particularly during nighttime episodes. Similar

benefits are observed in other neurological populations, such as Parkinson's disease, where exercise improves motor symptoms and quality of life [34], and fibromyalgia, where it reduces pain and fatigue while enhancing function [35]. Prior RLS research also shows that regular activity reduces symptom severity and improves sleep [9,12,14]. Movement-based and individualised exercise programmes could complement pharmacological treatment and address psychological barriers, enhancing coping and motivation.

Psychological factors influencing everyday life with RLS

Psychological barriers emerge from limited understanding, inadequate follow-up, and fragmented continuity of care. Patients reported that these gaps reduced trust in healthcare and contributed to feelings of abandonment and hopelessness, which in turn influenced their willingness to seek and engage with care. This may, in severe cases, reinforce depression, which could be seen as both a consequence of symptom burden and a barrier to treatment engagement [3,4]. Living with RLS also impairs quality of life and sleep, leading to tiredness, cognitive difficulties, and limitations in everyday life [6,9,13,35]. From an interpretive perspective, these findings align with Drieschner et al.'s Integral Model of Treatment Motivation [36], which conceptualises motivation as influenced by six internal determinants: problem recognition, level of suffering, perceived external pressure, perceived costs, perceived suitability of treatment, and outcome expectancy. These factors shape a patient's willingness to engage in treatment. Importantly, the model highlights that motivation does not directly translate into treatment engagement. Instead, the relationship is partial, with other factors such as cognitive capacity, prior experiences, and external barriers affecting whether motivation leads to actual behaviour. In an RLS context, inconsistent outcomes of pharmacological and non-pharmacological strategies can reduce confidence in treatment and treatment adherence, reinforcing a negative cycle where discouragement reduces the likelihood of achieving positive outcomes. Therefore, it is essential for healthcare professionals to address these inconsistencies and foster an environment that encourages patient engagement by improving communication about treatment expectations and providing robust support throughout the treatment process. These findings further highlight the importance of improved follow-up and a more comprehensive care approach in primary care. Previous studies have shown that responsibility for follow-up is often unclear, contributing to patients feeling unsupported [37]. Flexible and regular follow-up, including remote options, strengthens continuity, supports self-management, and allows ongoing adjustments to care. Research in other chronic conditions, such as Parkinson's disease and multiple sclerosis, similarly indicates that structured and continuous follow-up improves adherence, monitoring, and patient confidence [38,39]. Active patient involvement in individualised care plans is essential for sustaining hope for the future by living with a chronic illness [40].

Social factors influencing everyday life with RLS

Social barriers limit patients' everyday lives by restricting social engagement and occupational performance. This result aligns with previous studies where patients with RLS reported challenges with prolonged sitting, travel, and fatigue, often requiring careful planning and adjustment [6,9]. One way to understand these social barriers could be through the lens of Type D personality, a trait characterised by negative affectivity and social inhibition, which may limit help-seeking and communication with healthcare providers, thereby compounding social limitations [41]. This may also imply a risk that individuals with such traits are less likely to be identified and adequately supported within healthcare, as their needs may remain unexpressed. However, interpreting these findings solely through the lens of individual personality traits may overlook the importance of enhanced access to support from a multidisciplinary team to more comprehensively address patients' life situations. Health literacy may also play a crucial role in mitigating these effects of social and psychological barriers by enabling effective symptom management, informed treatment decisions, and self-care [42,43]. Importantly, health literacy should be viewed not only as an individual capacity but also as a responsibility of healthcare systems, highlighting the need for tailored communication and proactive support for patients with RLS. Social facilitators, in turn, support contact with patient organisations which can provide emotional, informational, belonging and practical support, including guidance on self-care, medication, and sleep hygiene [6]. Understanding family members can offer both practical and emotional support, assisting with social activities and daily routines despite fluctuating symptoms. Patients

who employ strategies such as adapting routines, listening to bodily cues, planning rest, moving during sedentary activities, and adjusting their diet, while also adopting flexible work hours and carrying medication, can be seen to further support proactive symptom management. These strategies are reflected by research on social support in other chronic conditions [44], highlighting its potential to act as a buffer against stress, promote adaptive coping, and reinforce resilience in patients with RLS. From a clinical perspective, social support not only contributes to symptom management but also enhances the ability of individuals with RLS to cope with daily life, fostering greater confidence, autonomy, and a sense of control.

Although the BPS model has been criticised as broad and hard to apply [45], our findings show its practical value in understanding RLS. Engel emphasised the interplay of biological, psychological, and social factors in health outcomes [16,17] and criticised reductionistic medicine for overlooking patient experiences, highlighting the physiological impact of personal narratives. Clinically, in an RLS setting, this supports addressing biological factors through symptom monitoring and physical activity ("B"), psychological factors *via* coping strategies and motivation ("P"), and social factors through family or patient organisation support ("S"). Integrating all three approaches reduces barriers, strengthens autonomy, and fosters hope, emphasising the value of holistic care in RLS. Future research should investigate interventions addressing the identified BPS-related barriers, particularly those targeting sleep disturbances and care-related stress, such as cognitive behavioural therapy, and evaluate their effects on symptom severity and overall BPS functioning in RLS. Studies could follow patient experiences and RLS symptoms before, during, and after treatment to identify factors that enhance adherence, resilience, and long-term benefits.

Several limitations warrant consideration. First, participants were recruited from a nationwide patient organisation, which may mean they were more engaged in their RLS care than the general population. However, including 28 strategically selected participants with diverse ages, genders, education, symptom burdens, and treatment regimens from different parts of Sweden strengthens the credibility and transferability of the findings. Second, telephone interviews may have limited the ability to capture non-verbal cues, which could affect the depth of the data. However, this format may also have made participants feel more at ease in disclosing sensitive information. Third, augmentation was not systematically assessed in this study. Although augmentation was mentioned in some of the interviews, the number of participants reporting experiences of augmentation was not systematically identified or quantified. Therefore, the potential influence of augmentation on participants' experiences of living with RLS could not be fully considered. Finally, as only three of the authors conducted the interviews, there is a potential risk of limited variation in data collection. However, the small number of interviewers may have contributed to greater consistency across interviews. Furthermore, the involvement of the full interdisciplinary research team in the analysis, supported by ongoing discussions, helped to mitigate this risk and to strengthen dependability and confirmability.

Conclusion

Comprehensive RLS care should address the interconnected biological, psychological, and social barriers that influence patients' everyday lives, while also recognizing facilitators such as lifestyle adaptations, self-care strategies, and social support. An integrated biopsychosocial approach could support healthcare professionals to identify individual needs, understand the complexity of living with RLS, and develop holistic care plans in collaboration with patients. Such an approach may help patients to identify self-care strategies that are meaningful and feasible in their everyday lives and may provide a framework for more integrated and holistic RLS care.

Acknowledgements

The authors thank all the participants of this study.

Author contributions

CRediT: **Elzana Odzakovic**: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing; **Elin Anefur**: Conceptualization, Data curation, Formal analysis, Methodology, Writing – review & editing; **Sandra Öberg**: Methodology, Writing – review &

editing; **Malin Jakobsson**: Methodology, Writing – review & editing; **Maria Björk**: Methodology, Writing – review & editing; **Susanne Knutsson**: Methodology, Writing – review & editing; **Lise-Lotte Jonasson**: Data curation, Methodology, Writing – review & editing; **Mattias Georgsson**: Methodology, Writing – review & editing; **Martin Ulander**: Methodology, Writing – review & editing; **Bengt Fridlund**: Data curation, Formal analysis, Methodology, Writing – review & editing; **Jonas Lind**: Methodology, Writing – review & editing; **Anders Broström**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Familjen Kampradsstiftelse, Grant/Award number: 20223144; Forskningsrådet i sydöstra; Sverige, Grant/Award Number: FORSS Grant FORSS-969214.

ORCID

Elzana Odzakovic http://orcid.org/0000-0002-8163-5045
Sandra Öberg http://orcid.org/0000-0001-9100-2674
Malin Jakobsson http://orcid.org/0000-0002-7344-1515
Maria Björk http://orcid.org/0000-0001-6419-2417
Susanne Knutsson http://orcid.org/0009-0003-8659-8698
Martin Ulander http://orcid.org/0000-0002-3810-0432
Bengt Fridlund http://orcid.org/0000-0002-5412-9497
Anders Broström http://orcid.org/0000-0003-1884-5696

Data availability statement

The data that support the findings of this study are available from the corresponding author, E.O, upon reasonable request.

References

[1] Allen RP, Picchietti DL, Garcia-Borreguero D, et al. Restless legs syndrome/Willis-Ekbom disease diagnostic criteria: updated International Restless Legs Syndrome Study Group consensus criteria–history, rationale, description, and significance. Sleep Med. 2014;15(8):860–873. doi: 10.1016/j.sleep.2014.03.025.
[2] Broström A, Alimoradi Z, Lind J, et al. Worldwide estimation of restless legs syndrome: a systematic review and meta-analysis of prevalence in the general adult population. J Sleep Res. 2023;32(3):e13783. doi: 10.1111/jsr.13783.
[3] Kocakaya H, Say B. Evaluation of emotional dysregulation in patients with restless legs syndrome. Sleep Biol Rhythms. 2023;21(4):447–454. doi: 10.1007/s41105-023-00467-z.
[4] An T, Sun H, Yuan L, et al. Associations of anxiety and depression with restless leg syndrome: a systematic review and meta-analysis. Front Neurol. 2024;15:1366839. doi: 10.3389/fneur.2024.1366839.
[5] Winkelman JW, Berkowski JA, DelRosso LM, et al. Treatment of restless legs syndrome and periodic limb movement disorder: an American Academy of Sleep Medicine clinical practice guideline. J Clin Sleep Med. 2025;21(1):137–152. doi: 10.5664/jcsm.11390.
[6] Odzakovic E, Ingelsbo Petersson A, Lindholm Ericsson E, et al. Experiences of sleep problems, subsequent daytime consequences, and self-care activities used to improve sleep among patients with restless legs syndrome: a qualitative content analysis. J Res Nurs. 2025;31:17449871251384535. doi: 10.1177/17449871251384535.
[7] Vlasie A, Trifu SC, Lupuleac C, et al. Restless legs syndrome: an overview of pathophysiology, comorbidities and therapeutic approaches. Exp Ther Med. 2022;23(2):185. doi: 10.3892/etm.2021.11108.
[8] Kusnanto H, Agustian D, Hilmanto D. Biopsychosocial model of illnesses in primary care: a hermeneutic literature review. J Family Med Prim Care. 2018 ;7(3):497–500. doi: 10.4103/jfmpc.jfmpc_145_17.
[9] Odzakovic E, Allgurin M, Jonasson LL, et al. Experiences of facilitators and barriers for fulfilment of human needs when living with restless legs syndrome: a qualitative study. Int J Qual Stud Health Well-Being. 2024;19(1):2348884. doi: 10.1080/17482631.2024.2348884.
[10] Holzknecht E, Hochleitner M, Wenning GK, et al. Gender differences in clinical, laboratory and polysomnographic features of restless legs syndrome. J Sleep Res. 2020;29(3):e12875. doi: 10.1111/jsr.12875.

[11] Kim TJ, Kim MH, Kim JH, et al. Change of iron content in brain regions after intravenous iron therapy in restless legs syndrome: quantitative susceptibility mapping study. Sleep. 2023;46(8):zsad154. doi: 10.1093/sleep/zsad154.
[12] Harrison EG, Keating JL, Morgan PE. Non-pharmacological interventions for restless legs syndrome: a systematic review of randomised controlled trials. Disabil Rehabil. 2019;41(17):2006–2014. doi: 10.1080/09638288.2018.1453875.
[13] Broström A, Alimoradi Z, Odzakovic E, et al. Quality of life among patients with restless legs syndrome: a systematic review and meta-analysis. J Clin Neurosci. 2024;122:80–91. doi: 10.1016/j.jocn.2024.02.027.
[14] Odzakovic E, Sandlund C, Hellström A, et al. Self-care behaviours in patients with restless legs syndrome (RLS): development and psychometric testing of the RLS-self-care behaviour questionnaire. J Sleep Res. 2025;34(3):e14390.
[15] Odzakovic E, Eliasson A, Jansson P, et al. Prerequisites for self-care actions in individuals with restless legs syndrome—a deductive qualitative analysis based on the COM-B model. J Health Psychol. 2025;30(13):4059–4074. doi: 10.1177/13591053251315379.
[16] Engel GL. The need for a new medical model: a challenge for biomedicine. Science. 1977;196(4286):129–136. doi: 10.1126/science.847460.
[17] Engel GL. The clinical application of the biopsychosocial model. Am J Psychiatry. 1980;137(5):535–544. doi: 10.1176/ajp.137.5.535.
[18] Fricchione G. Mind body medicine: a modern bio-psycho-social model forty-five years after Engel. Biopsychosoc Med. 2023;17(1):12. doi: 10.1186/s13030-023-00268-3.
[19] Dong HJ, Bäckryd E. Teaching the biopsychosocial model of chronic pain: whom are we talking to? Patient Educ Couns. 2023;110:107645. doi: 10.1016/j.pec.2023.107645.
[20] Gibson G. What can the treatment of Parkinson's disease learn from dementia care: applying a bio-psycho-social approach to Parkinson's disease. Int J Older People Nurs. 2017;12(4). doi: 10.1111/opn.12159.
[21] Ormstad H, Eilertsen G. A biopsychosocial model of fatigue and depression following stroke. Med Hypotheses. 2015;85(6):835–841. doi: 10.1016/j.mehy.2015.10.001.
[22] Graneheim UH, Lundman B. Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. Nurse Educ Tod. 2004;24(2):105–112. doi: 10.1016/j.nedt.2003.10.001.
[23] Hellström A, Pakpour A, Odzakovic E, et al. The psychometric properties of the Pearlin Mastery Scale in persons living with restless legs syndrome. PLoS One. 2024;19(10):e0311259. doi: 10.1371/journal.pone.0311259.
[24] Moser A, Korstjens I. Series: practical guidance to qualitative research. Part 3: sampling, data collection and analysis. Eur J Gen Pract. 2018 ;24(1):9–18. doi: 10.1080/13814788.2017.1375091.
[25] Lincoln YS, Guba EG. Naturalistic inquiry. Newbury Park: Sage Publications; 1985.
[26] World Medical Association. Declaration of Helsinki: ethical principles for medical research involving human subjects; 2024.
[27] Schormair B, Zhao C, Bell S, et al. Genome-wide meta-analyses of restless legs syndrome yield insights into genetic architecture, disease biology and risk prediction. Nat Genet. 2024 ;56(6):1090–1099. doi: 10.1038/s41588-024-01763-1.
[28] Earley CJ, Jones BC, Ferré S. Brain-iron deficiency models of restless legs syndrome. Exp Neurol. 2022 ;356:114158. doi: 10.1016/j.expneurol.2022.114158.
[29] Ellis JG, Perlis ML, Espie CA, et al. The natural history of insomnia: predisposing, precipitating, coping, and perpetuating factors over the early developmental course of insomnia. Sleep. 2021;44(9):zsab095. doi: 10.1093/sleep/zsab095.
[30] Wright CD, Tiani AG, Billingsley AL, et al. A framework for understanding the role of psychological processes in disease development, maintenance, and treatment: the 3P-disease model. Front Psychol. 2019;10:2498. doi: 10.3389/fpsyg.2019.02498.
[31] Brand S, Lehtinen A, Hatzinger M, et al. Comparison of sleep EEG profiles of patients suffering from restless legs syndrome, restless legs syndrome and depressive symptoms, and major depressive disorders. Neuropsychobiology. 2010;61(1):41–48. doi: 10.1159/000262179.
[32] Benbir Senel G, Tunali A, Demirel O, et al. Spectral EEG and heart rate changes associated with leg movements during the suggested immobilisation test in patients with restless legs syndrome. J Sleep Res. 2025 ;34(3):e14394. doi: 10.1111/jsr.14394.
[33] Karroum EG, Bachmann CG, Bronstone A, et al. Efficacy and safety of tonic motor activation (TOMAC) for restless legs syndrome as adjunctive and monotherapy: an individual participant data systematic review and meta-analysis. Sleep Med. 2026;140:108810. doi: 10.1016/j.sleep.2026.108810.
[34] Bispo DPCF, Lins CCSA, Hawkes KL, et al. The positive effects of physical activity on quality of life in Parkinson's disease: a systematic review. Geriatrics (Basel). 2024;9(4):94. doi: 10.3390/geriatrics9040094.
[35] Taylor TL, Dodds F, Tharpe M, et al. The potential impact of exercise on affect and neuroinflammation in older adults living with fibromyalgia: a scoping review. Front Hum Neurosci. 18:6. doi: 10.3389/fnhum.2024.1463935.
[36] Drieschner KH, Lammers SM, van der Staak CPF. Treatment motivation: an attempt for clarification of an ambiguous concept. Clin Psychol Rev. 2004;23(8):1115–1137. doi: 10.1016/j.cpr.2003.09.003.
[37] Khazen M, Shalev L, Golan-Cohen A, et al. Teamwork among primary care staff to achieve regular follow-up of chronic patients. Ann Fam Med. 2025;23(2):100–107. doi: 10.1370/afm.240176.

[38] Weise D, Claus I, Dresel C, et al. Multidisciplinary care in Parkinson's disease. J Neural Transm (Vienna). 2024;131(10):1217–1227. doi: 10.1007/s00702-024-02807-w.
[39] Beczek A, Landt EM, Storr LK, et al. Barriers to clinical follow-up visits in multiple sclerosis: a nationwide register-based study. Mult Scler J Exp Transl Clin. 2024;10(4):20552173241307619. doi: 10.1177/20552173241307619.
[40] Benzein E, Norberg A, Saveman BI. The meaning of the lived experience of hope in patients with cancer in palliative home care. Palliat Med. 1998;12:117–126.
[41] Mols F, Denollet J. Type D personality in the general population: a systematic review. J Psychosom Res. 2010;69:567–575. doi: 10.1016/j.jpsychores.2010.07.007.
[42] Georgsson M, Odzakovic E, Björk M, et al. Validation of the eHealth Literacy Scale instrument in a restless legs syndrome population: classical test theory and Rasch analysis study. J Med Internet Res. 2025;27:e68474. doi: 10.2196/68474.
[43] Säwén A, Odzakovic E, Fridlund B, et al. Factors associated with shared decision-making and decisional conflicts among patients with restless legs syndrome: a cross-sectional study. Asian J Soc Health Behav. 2026;9(3):166–176. doi: 10.4103/shb.shb_388_25.
[44] Cohen S, Wills TA. Stress, social support, and the buffering hypothesis. Psychol Bull. 1985;98(2):310–357. doi: 10.1037/0033-2909.98.2.310.
[45] Lane RD. Is it possible to bridge the biopsychosocial and biomedical models? Biopsychosoc Med. 2014;8(1):3. doi: 10.1186/1751-0759.